

	(T_M, S_M, N_S)		(T_P, S_P, N_S)		(T_L, S_L, N_S)
	I_{KD}	I_G	I_R	I_P	I_{Luk}
Commutativity of $\wedge, \vee$	✓	✓	✓	✓	✓
Associativity of $\wedge, \vee$	✓	✓	✓	✓	✓
Distributivity of $\wedge$ over $\vee$	✓	✓			
Distributivity of $\vee$ over $\wedge$	✓	✓			
Distrib. of $\rightarrow$ over $\vee, \wedge$	✓	✓			
Double negation $\neg\neg p = p$	✓	✓	✓	✓	✓
Law of excluded middle					✓
Law of non contradiction					✓
De Morgan's laws	✓	✓	✓	✓	✓
Material Implication	✓		✓		✓
Contraposition	✓		✓		✓

Table B.6: Common properties for different configurations

Following are other common aggregators:

$$A_M(x_1, \dots, x_n) = \frac{1}{n} \sum_{i=1}^n x_i \quad (\text{mean})$$

$$A_{pM}(x_1, \dots, x_n) = \left(\frac{1}{n} \sum_{i=1}^n x_i^p \right)^{\frac{1}{p}} \quad (\text{p-mean})$$

$$A_{pME}(x_1, \dots, x_n) = 1 - \left(\frac{1}{n} \sum_{i=1}^n (1 - x_i)^p \right)^{\frac{1}{p}} \quad (\text{p-mean error})$$

Where A_{pM} is the generalized mean, and A_{pME} can be understood as the generalized mean measured w.r.t. the errors. That is, A_{pME} measures the power of the deviation of each value from the ground truth 1. A few particular values of p yield special cases of aggregators. Notably:

- $\lim_{p \rightarrow +\infty} A_{pM}(x_1, \dots, x_n) = \max(x_1, \dots, x_n),$
- $\lim_{p \rightarrow -\infty} A_{pM}(x_1, \dots, x_n) = \min(x_1, \dots, x_n),$
- $\lim_{p \rightarrow +\infty} A_{pME}(x_1, \dots, x_n) = \min(x_1, \dots, x_n),$
- $\lim_{p \rightarrow -\infty} A_{pME}(x_1, \dots, x_n) = \max(x_1, \dots, x_n).$

These "smooth" min (resp. max) approximators are good candidates for $\forall$ (resp. $\exists$) in a fuzzy context. The value of p leaves more or less room for outliers depending on the use case and its needs. Note that A_{pME} and A_{pM} are related in the same way that $\exists$ and $\forall$ are related using the definition $\exists \equiv \neg\forall\neg$, where $\neg$ would be approximated by the standard negation.

We propose to use A_{pME} with $p \geq 1$ to approximate $\forall$ and A_{pM} with $p \geq 1$ to approximate $\exists$. When $p \geq 1$, these operators resemble the l^p norm of a vector $u = (u_1, u_2, \dots, u_n)$, where $\|u\|_p = (|u_1|^p + |u_2|^p + \dots + |u_n|^p)^{1/p}$. In our case, many properties of the l^p norm can apply to A_{pM} (positive homogeneity, triangular inequality, ...).

Appendix C. Analyzing Gradients of Generalized Mean Aggregators

[69] show that some operators used in Fuzzy Logics are unsuitable for use in a differentiable learning setting. Three types of gradient problems commonly arise in fuzzy logic operators.

Single-Passing The derivatives of some operators are non-null for only one argument. The gradients propagate to only one input at a time.

Vanishing Gradients The gradients vanish on some part of the domain. The learning does not update inputs that are in the vanishing domain.

Exploding Gradients Large error gradients accumulate and result in unstable updates.

Tables C.7 and C.8 summarize their conclusions for the most common operators. Also, we underline here exploding gradients issues that arise experimentally in A_{pM} and A_{pME} , which are not in the original report. Given the truth values of n propositions $(x_1, \dots, x_n)$ in $[0, 1]^n$:

$$1. A_{pM}(x_1, \dots, x_n) = \left(\frac{1}{n} \sum_i x_i^p \right)^{\frac{1}{p}}$$

The partial derivatives are $\frac{\partial A_{pM}(x_1, \dots, x_n)}{\partial x_i} = \frac{1}{n} \frac{1}{p} \left(\sum_{j=1}^n x_j^p \right)^{\frac{1}{p}-1} x_i^{p-1}$.

When $p > 1$, the operator weights more for inputs with a higher true value –i.e. their partial derivative is also higher – and suits for existential quantification. When $p < 1$, the operator weights more for inputs with a lower true value and suits for universal quantification.

Exploding Gradients When $p > 1$, if $\sum_{j=1}^n x_j^p \rightarrow 0$, then $\left(\sum_{j=1}^n x_j^p \right)^{\frac{1}{p}-1} \rightarrow \infty$ and the gradients explode. When $p < 1$, if $x_i \rightarrow 0$, then $x_i^{p-1} \rightarrow \infty$.

$$2. A_{pME}(x_1, \dots, x_n) = 1 - \left(\frac{1}{n} \sum_i (1 - x_i)^p \right)^{\frac{1}{p}}$$

The partial derivatives are $\frac{\partial A_{pME}(x_1, \dots, x_n)}{\partial x_i} = \frac{1}{n} \frac{1}{p} \left(\sum_{j=1}^n (1 - x_j)^p \right)^{\frac{1}{p}-1} (1 - x_i)^{p-1}$. When $p > 1$, the operator weights more for inputs with a lower true value –i.e. their partial derivative is also higher – and suits for universal quantification. When $p < 1$, the operator weights more for inputs with a higher true value and suits for existential quantification.

Exploding Gradients

When $p > 1$, if $\sum_{j=1}^n (1 - x_j)^p \rightarrow 0$, then $\left(\sum_{j=1}^n (1 - x_j)^p \right)^{\frac{1}{p}-1} \rightarrow \infty$ and the gradients explode. When $p < 1$, if $1 - x_i \rightarrow 0$, then $(1 - x_i)^{p-1} \rightarrow \infty$.

We propose the following *stable product configuration* that does not have any of the aforemen-

	Single-Passing	Vanishing	Exploding
<i>Goedel (mininum)</i>			
T_M, S_M	$\mathbf{\times}$		
I_{KD}	$\mathbf{\times}$		
I_G	$\mathbf{\times}$	$\mathbf{\times}$	
<i>Goguen (product)</i>			
T_P, S_P		$(\mathbf{\times})$	
I_R		$(\mathbf{\times})$	
I_{KD}		$\mathbf{\times}$	$(\mathbf{\times})$
<i>Łukasiewicz</i>			
T_L, S_L		$\mathbf{\times}$	
I_{Luk}		$\mathbf{\times}$	

Table C.7: Gradient problems for some binary connectives. $(\mathbf{\times})$ means that the problem only appears on an edge case.

	Single-Passing	Vanishing	Exploding
A_{T_M} / A_{S_M}	$\mathbf{\times}$		
A_{T_P} / A_{S_P}		$\mathbf{\times}$	
A_{T_L} / A_{S_L}		$\mathbf{\times}$	
A_{pM}			$(\mathbf{\times})$
A_{pME}			$(\mathbf{\times})$

Table C.8: Gradient problems for some aggregators. $(\mathbf{\times})$ means that the problem only appears on an edge case.

tioned gradient problems:

$$\pi_0(x) = (1 - \epsilon)x + \epsilon \quad (\text{C.1})$$

$$\pi_1(x) = (1 - \epsilon)x \quad (\text{C.2})$$

$$N_S(x) = 1 - x \quad (\text{C.3})$$

$$T'_P(x, y) = \pi_0(x)\pi_0(y) \quad (\text{C.4})$$

$$S'_P(x, y) = \pi_1(x) + \pi_1(y) - \pi_1(x)\pi_1(y) \quad (\text{C.5})$$

$$I'_R(x, y) = 1 - \pi_0(x) + \pi_0(x)\pi_1(y) \quad (\text{C.6})$$

$$A'_{pM}(x_1, \dots, x_n) = \left(\frac{1}{n} \sum_{i=1}^n \pi_0(x_i)^p \right)^{\frac{1}{p}} \quad p \geq 1 \quad (\text{C.7})$$

$$A'_{pME}(x_1, \dots, y_n) = 1 - \left(\frac{1}{n} \sum_{i=1}^n (1 - \pi_1(x_i))^p \right)^{\frac{1}{p}} \quad p \geq 1 \quad (\text{C.8})$$

N_S is the operator for negation, T'_P for conjunction, S'_P for disjunction, I'_P for implication, A'_{pM} for existential aggregation, A'_{pME} for universal aggregation.